Problem 1.14 [Difficulty 1]

1.14 The mean free path λ of a molecule of gas is the average distance it travels before collision with another molecule. It is given by

$$\lambda = C\frac{m}{\rho d^2}$$

where m and d are the molecule's mass and diameter, respectively, and ρ is the gas density. Determine the dimensions of constant C for a dimensionally consistent equation.

Given: Equation for mean free path of a molecule.

Find: Dimensions of C for a diemsionally consistent equation.

Solution: Use the mean free path equation. Then "solve" for C and use dimensions.

The mean free path equation is $\lambda = C\cdot\frac{m}{\rho\cdot d^2}$

"Solving" for C, and using dimensions $C = \frac{\lambda\cdot\rho\cdot d^2}{m}$

$$C = \frac{L\times\frac{M}{L^3}\times L^2}{M} = 0$$

The constant C is dimensionless.

Problem 1.17

[Difficulty: 1]

1.17 Derive the following conversion factors:

(a) Convert a pressure of 1 psi to kPa.

(b) Convert a volume of 1 liter to gallons.

(c) Convert a viscosity of 1 $lbf \cdot s/ft^2$ to $N \cdot s/m^2$.

Given: Pressure, volume and density data in certain units

Find: Convert to different units

Solution:

Using data from tables

(a) $$1 \cdot psi = 1 \cdot psi \times \frac{6895 \cdot Pa}{1 \cdot psi} \times \frac{1 \cdot kPa}{1000 \cdot Pa} = 6.89 \cdot kPa$$

(b) $$1 \cdot liter = 1 \cdot liter \times \frac{1 \cdot quart}{0.946\, liter} \times \frac{1 \cdot gal}{4 \cdot quart} = 0.264\, gal$$

(c) $$1 \cdot \frac{lbf \cdot s}{ft^2} = 1 \cdot \frac{lbf \cdot s}{ft^2} \times \frac{4.448\, N}{1 \cdot lbf} \times \left(\frac{\frac{1}{12} \cdot ft}{0.0254\, m} \right)^2 = 47.9 \cdot \frac{N \cdot s}{m^2}$$

Problem 1.18

[Difficulty: 1]

1.18 Express the following in SI units:

(a) 100 cfm (ft^3/min)
(b) 5 gal
(c) 65 mph
(d) 5.4 acres

Given: Quantities in English Engineering (or customary) units.

Find: Quantities in SI units.

Solution: Use Table and other sources (e.g., Google)

(a) $$100\cdot\frac{ft^3}{m} = 100\cdot\frac{ft^3}{min}\times\left(\frac{0.0254\cdot m}{1\cdot in}\times\frac{12\cdot in}{1\cdot ft}\right)^3\times\frac{1\cdot min}{60\cdot s} = 0.0472\cdot\frac{m^3}{s}$$

(b) $$5\cdot gal = 5\cdot gal\times\frac{231\cdot in^3}{1\cdot gal}\times\left(\frac{0.0254\cdot m}{1\cdot in}\right)^3 = 0.0189\cdot m^3$$

(c) $$65\cdot mph = 65\cdot\frac{mile}{hr}\times\frac{1852\cdot m}{1\cdot mile}\times\frac{1\cdot hr}{3600\cdot s} = 29.1\cdot\frac{m}{s}$$

(d) $$5.4\cdot acres = 5.4\cdot acre\times\frac{4047\cdot m^3}{1\cdot acre} = 2.19\times 10^4\cdot m^2$$

Problem 1.20 [Difficulty 1

1.20 Derive the conversion factors for the following quantities for volume flow rate
(a) Converting in^3/min to mm^3/s.
(b) Converting gallons per minute (gpm) to m^3/s.
(c) Converting gpm to liters/min.
(d) Converting cubic feet per minute (cfm) to m^3/s.

Given: Data in given units

Find: Convert to different units

Solution:

(a) $$1\cdot\frac{in^3}{min} = 1\cdot\frac{in^3}{min}\times\left(\frac{0.0254\,m}{1\cdot in}\times\frac{1000\,mm}{1\cdot m}\right)^3\times\frac{1\cdot min}{60\cdot s} = 273\cdot\frac{mm^3}{s}$$

(b) $$1\cdot\frac{m^3}{s} = 1\cdot\frac{m^3}{s}\times\frac{1\cdot gal}{4\times 0.000946\,m^3}\times\frac{60\cdot s}{1\cdot min} = 15850\,gpm$$

(c) $$1\cdot\frac{liter}{min} = 1\cdot\frac{liter}{min}\times\frac{1\cdot gal}{4\times 0.946\cdot liter}\times\frac{60\cdot s}{1\cdot min} = 0.264\cdot gpm$$

(d) $$1\cdot SCFM = 1\cdot\frac{ft^3}{min}\times\left(\frac{0.0254\cdot m}{\frac{1}{12}\cdot ft}\right)^3\times\frac{60\cdot min}{1\cdot hr} = 1.70\cdot\frac{m^3}{hr}$$

Problem 1.24

[Difficulty: 3]

1.24 From Appendix A, the viscosity μ ($N \cdot s/m^2$) of water at temperature T (K) can be computed from $\mu = A10^{B/(T-C)}$, where $A = 2.414 \times 10^{-5}$ $N \cdot s/m^2$, $B = 247.8$ K, and $C = 140$ K. Determine the viscosity of water at 30°C, and estimate its uncertainty if the uncertainty in temperature measurement is ±0.5°C.

Given: Data on water

Find: Viscosity; Uncertainty in viscosity

Solution:

The data is: $A = 2.414 \times 10^{-5} \cdot \frac{N \cdot s}{m^2}$ $\quad B = 247.8 \cdot K$ $\quad C = 140 \cdot K$ $\quad T = 303 \cdot K$

The uncertainty in temperature is $u_T = \frac{0.5 \cdot K}{293 \cdot K}$ $\quad u_T = 0.171 \cdot \%$

Also $\mu(T) = A \cdot 10^{\frac{B}{(T-C)}}$ $\quad$ Evaluating $\mu(293 \cdot K) = 1.005 \times 10^{-3} \cdot \frac{N \cdot s}{m^2}$

For the uncertainty

$$\frac{d}{dT}\mu(T) = -\frac{A \cdot B \cdot \ln(10)}{10^{\frac{B}{C-T}} \cdot (C - T)^2}$$

Hence $u_\mu(T) = \left| \frac{T}{\mu(T)} \cdot \frac{d}{dT}\mu(T) \cdot u_T \right| = \frac{\ln(10) \cdot \left| B \cdot T \cdot u_T \right|}{\left(\left| C - T \right| \right)^2}$ $\quad$ Evaluating $u_\mu(T) = 1.11 \cdot \%$

Problem 2.17

[Difficulty: 2]

2.17 The variation with temperature of the viscosity of air is represented well by the empirical Sutherland correlation

$$\mu = \frac{bT^{1/2}}{1 + S/T}$$

Best-fit values of b and S are given in Appendix A. Develop an equation in SI units for kinematic viscosity versus temperature for air at atmospheric pressure. Assume ideal gas behavior. Check by using the equation to compute the kinematic viscosity of air at 0°C and at 100°C and comparing to the data in Appendix A (Table A.10); plot the kinematic viscosity for a temperature range of 0°C to 100°C, using the equation and the data in Table A.10.

Given: Sutherland equation

Find: Corresponding equation for kinematic viscosity

Solution:

Governing equation: $\mu = \dfrac{b \cdot T^{\frac{1}{2}}}{1 + \frac{S}{T}}$ Sutherland equation $\quad p = \rho \cdot R \cdot T$ Ideal gas equation

Assumptions: Sutherland equation is valid; air is an ideal gas

The given data is $\quad b = 1.458 \times 10^{-6} \cdot \dfrac{kg}{m \cdot s \cdot K^{\frac{1}{2}}} \quad S = 110.4 \cdot K \quad R = 286.9 \cdot \dfrac{J}{kg \cdot K} \quad p = 101.3 \cdot kPa$

The kinematic viscosity is

$$\nu = \frac{\mu}{\rho} = \frac{\mu \cdot R \cdot T}{p} = \frac{R \cdot T}{p} \cdot \frac{b \cdot T^{\frac{1}{2}}}{1 + \frac{S}{T}} = \frac{R \cdot b}{p} \cdot \frac{T^{\frac{3}{2}}}{1 + \frac{S}{T}} = \frac{b' \cdot T^{\frac{3}{2}}}{1 + \frac{S}{T}}$$

where $\quad b' = \dfrac{R \cdot b}{p} \qquad b' = 4.129 \times 10^{-9} \dfrac{m^2}{K^{1.5} \cdot s}$

$$b' = 286.9 \cdot \frac{N \cdot m}{kg \cdot K} \times 1.458 \times 10^{-6} \cdot \frac{kg}{m \cdot s \cdot K^{\frac{1}{2}}} \times \frac{m^2}{101.3 \times 10^3 \cdot N} = 4.129 \times 10^{-9} \cdot \frac{m^2}{s \cdot K^{\frac{3}{2}}}$$

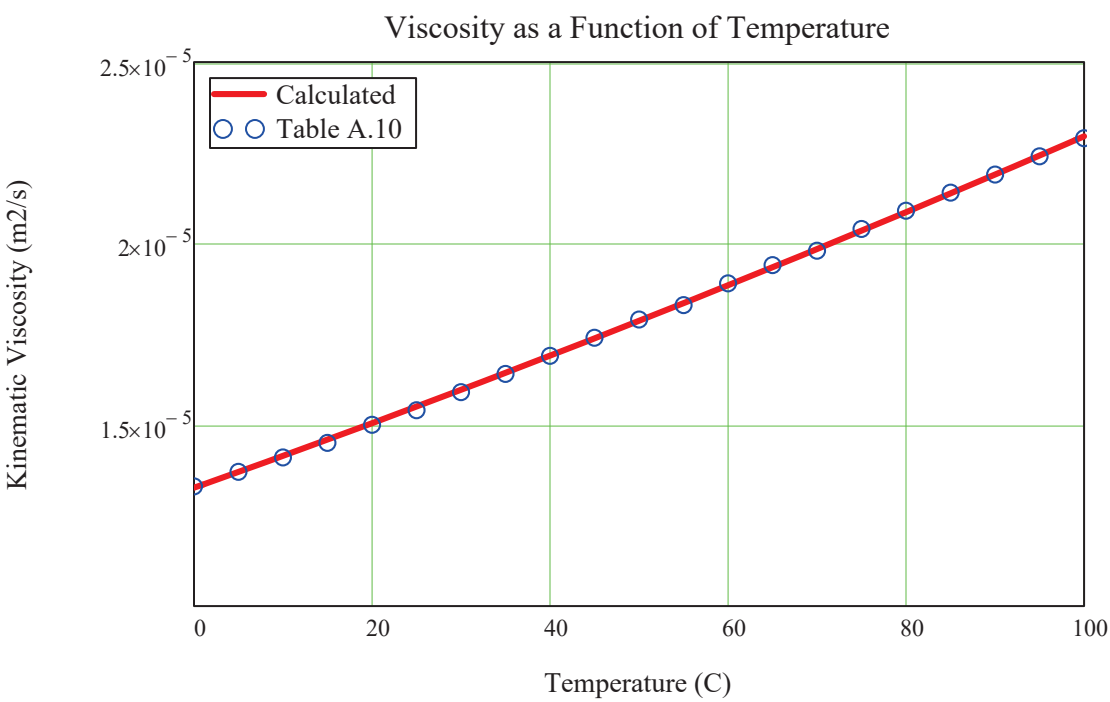
Hence $\quad \nu = \dfrac{b' \cdot T^{\frac{3}{2}}}{1 + \frac{S}{T}}$ with $b' = 4.129 \times 10^{-9} \cdot \dfrac{m^2}{s \cdot K^{\frac{3}{2}}} \qquad S = 110.4\ K$

Check with Appendix A, Table A.10. At $T = 0\,°C$ we find $\quad T = 273.1\,K \quad \nu = 1.33 \times 10^{-5} \cdot \frac{m^2}{s}$

$$\nu = \frac{4.129 \times 10^{-9} \frac{m^2}{s \cdot K^{\frac{3}{2}}} \times (273.1 \cdot K)^{\frac{3}{2}}}{1 + \frac{110.4}{273.1}} \qquad \nu = 1.33 \times 10^{-5} \cdot \frac{m^2}{s} \qquad \text{Check!}$$

At $T = 100\,°C$ we find $\quad T = 373.1\,K \quad \nu = 2.29 \times 10^{-5} \cdot \frac{m^2}{s}$

$$\nu = \frac{4.129 \times 10^{-9} \frac{m^2}{s \cdot K^{\frac{3}{2}}} \times (373.1 \cdot K)^{\frac{3}{2}}}{1 + \frac{110.4}{373.1}} \qquad \nu = 2.30 \times 10^{-5} \cdot \frac{m^2}{s} \qquad \text{Check!}$$



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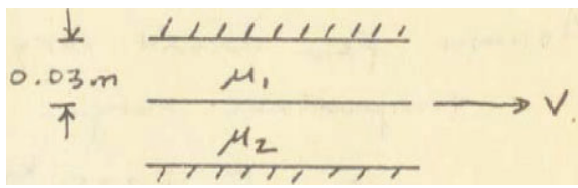
Problem 2.20

(Difficulty: 2)

2.20 A very large thin plate is centered in a gap of width $0.06\ m$ with different oils of unknown viscosities above and below; one viscosity is twice the other. When the plate is pulled at a velocity of $0.3\ \frac{m}{s}$, the resulting force on one square meter of plate due to the viscous shear on both sides is $29\ N$.. Assuming viscous flow and neglecting all end effects, calculate the viscosities of the oils.



Given: Viscosity: $\mu_2 = 2\mu_1$. Width of gap: $h = 0.06\ m$. Velocity:$V = 0.3\ \frac{m}{s}$. Force per square meter: $F = 29\ \frac{N}{m^2}$

Find: μ_1 and μ_2

Assumption: Viscous flow with linear velocity profiles, negligible end effects.

Solution: Use Newton's law relating shear stress to viscosity and velocity gradient. The relation between the two viscosities is

$$\mu_2 = 2\mu_1$$

Because the gaps are equal and the plate velocity is the same for both fluids, the velocity gradient is the same for both sides of the plate:

$$\frac{dV}{dy} = \frac{V}{0.5h} = \frac{0.3\ \frac{m}{s}.}{0.5 \times 0.06\ m} = 10\ \frac{1}{s}$$

For a Newtonian fluid with a linear velocity profile, we have

$$\tau = \mu\frac{dV}{dy} = \mu\frac{\Delta V}{\Delta y}$$

The force on the plate due to the top layer of fluid is

$$\tau_1 = \mu_1\frac{\Delta V}{\Delta y} = \mu_1\frac{0.3\frac{m}{s}}{0.03\ m} = \mu_1 101$$

Similarly, the force on bottom of the plate is

$$\tau_2 = \mu_2\frac{\Delta V}{\Delta y} = \mu_2\frac{0.3\frac{m}{s}}{0.03\ m} = \mu_2 10\frac{1}{s}$$

The total force per unit area equals the sum of the two shear stresses, where for the 1 m^2 plate the shear stress is equal to 29 N/m^2.

$$\frac{F}{A} = \tau_1 + \tau_2 = 29\ \frac{N}{m^2}$$

Or, since μ_2 = times μ_1

$$\mu_1 101 + \mu_2 10 \frac{1}{s} = 3\mu_1 10 \frac{1}{s} = 29\ \frac{N}{m^2}$$

$$\mu_1 = \frac{1}{3} \times 29\ \frac{N}{m^2} \times \frac{s}{10} = 0.967 \frac{N \cdot s}{m^2} = 0.967\ Pa \cdot s$$

$$\mu_2 = 2\mu_1 = 1.934\ Pa \cdot s$$